

Norway

SLEIPNER

Volve F

F-1

F-1

Design: F-1

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
Design:	F-1	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9		
Site Position:		Northing:	6,478,563.53 m
From:	Map	Easting:	435,050.02 m
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in
		Latitude:	58° 26' 29.8068 N
		Longitude:	1° 53' 14.9285 E
		Grid Convergence:	-0.95 °

Well	F-1		
Well Position	+N/-S	3.17 m	Northing:
	+E/-W	-3.53 m	Easting:
Position Uncertainty	0.00 m		Wellhead Depth:
			91.00 m
			Latitude:
			58° 26' 29.9072 N
			Longitude:
			1° 53' 14.7075 E
			Water Depth:
			91.00 m

Wellbore	F-1		
Magnetics	Model Name	Sample Date	Declination
			(°)
			Dip Angle
			(°)
			Field Strength
			(nT)
	MNETICREFERENCE	02.11.2005	-2.54
			71.62
			50,284

Design	F-1		
Audit Notes:			
Version:	1.0	Phase:	ACTUAL
		Tie On Depth:	145.90
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(m)	(m)	(m)
	145.90	3.17	-3.53
			Direction
			(°)
			302.24

Survey Program	Date	30.09.2017		
From	To	Survey (Wellbore)	Tool Name	Description
(m)	(m)			
150.15	210.70	15/9 F1 36" MWD (F-1)	Magnetic, old	Magnetic Tools (MWD, EMS)
237.60	358.50	15/9 F1 26" Gyro Scientific Modular (F-1)	NOT_useMWD gyro, Baker Hughes' MWD gyro service (GyroTrak)	
378.00	737.40	15/9 F-1 26" SDI Keeper Gyro Single Shot	Gyro, old	Other Gyro Tools
756.80	1,335.60	15/9 F-1 26" OnTrak (F-1)	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)
1,387.10	2,587.00	15/9 F-1 17 1/2" AutoTrak G3 (F-1)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
2,653.20	3,620.30	15/9 F-1 8 1/2" AutoTrak G3 (F-1)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)

Survey										
Measured	Inclination	Azimuth	Vertical	+N/-S	+E/-W	Vertical	Dogleg	Build	Turn	
Depth	(°)	(°)	Depth	(m)	(m)	Section	Rate	Rate	Rate	
(m)			(m)			(m)	(°/30m)	(°/30m)	(°/30m)	
145.90	0.00	0.00	145.90	3.17	-3.53	0.00	0.000	0.000	0.000	
150.15	0.00	0.00	150.15	3.17	-3.53	0.00	0.000	0.000	0.000	
160.45	0.00	0.00	160.45	3.17	-3.53	0.00	0.000	0.000	0.000	
170.04	0.00	0.00	170.04	3.17	-3.53	0.00	0.000	0.000	0.000	
179.79	0.00	0.00	179.79	3.17	-3.53	0.00	0.000	0.000	0.000	
189.74	0.00	0.00	189.74	3.17	-3.53	0.00	0.000	0.000	0.000	

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Well:	F-1	North Reference:	Grid
Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
Design:	F-1	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
198.50	0.00	0.00	198.50	3.17	-3.53	0.00	0.000	0.000	0.000	
210.70	0.00	0.00	210.70	3.17	-3.53	0.00	0.000	0.000	0.000	
237.60	1.84	302.24	237.60	3.40	-3.90	0.43	2.052	2.052	0.000	
251.10	1.91	304.70	251.09	3.64	-4.27	0.87	0.237	0.156	5.467	
264.90	2.47	305.76	264.88	3.94	-4.70	1.40	1.221	1.217	2.304	
279.20	2.05	306.81	279.17	4.28	-5.15	1.96	0.885	-0.881	2.203	
292.70	1.80	304.00	292.66	4.54	-5.52	2.42	0.594	-0.556	-6.244	
302.80	1.13	330.72	302.76	4.72	-5.70	2.66	2.791	-1.990	79.366	
315.50	1.41	333.18	315.45	4.96	-5.83	2.91	0.674	0.661	5.811	
332.20	2.40	314.19	332.14	5.39	-6.18	3.42	2.086	1.778	-34.114	
342.20	2.78	301.18	342.13	5.66	-6.54	3.87	2.093	1.140	-39.030	
358.50	3.30	293.10	358.41	6.05	-7.30	4.73	1.238	0.957	-14.871	
378.00	3.07	298.60	377.88	6.52	-8.28	5.80	0.588	-0.354	8.462	
414.80	3.30	293.64	414.62	7.42	-10.11	7.84	0.292	0.187	-4.043	
455.40	3.07	292.72	455.16	8.31	-12.19	10.06	0.174	-0.170	-0.680	
495.40	2.84	290.41	495.11	9.07	-14.10	12.09	0.194	-0.172	-1.732	
535.60	2.77	293.07	535.26	9.79	-15.93	14.02	0.110	-0.052	1.985	
575.80	3.06	287.04	575.41	10.49	-17.85	16.02	0.315	0.216	-4.500	
616.50	4.01	305.32	616.03	11.63	-20.05	18.49	1.078	0.700	13.474	
656.80	6.13	317.80	656.17	14.04	-22.65	21.97	1.770	1.578	9.290	
697.10	8.24	324.71	696.15	17.99	-25.76	26.71	1.695	1.571	5.144	
720.00	9.57	322.37	718.78	20.84	-27.87	30.01	1.805	1.742	-3.066	
737.40	10.71	320.81	735.91	23.24	-29.78	32.90	2.022	1.966	-2.690	
756.80	10.39	320.18	754.98	25.98	-32.04	36.28	0.526	-0.495	-0.974	
798.90	9.88	320.50	796.42	31.68	-36.76	43.32	0.366	-0.363	0.228	
838.30	10.04	315.72	835.23	36.75	-41.31	49.87	0.641	0.122	-3.640	
868.00	11.16	309.73	864.42	40.44	-45.33	55.24	1.586	1.131	-6.051	
878.20	11.28	310.41	874.43	41.72	-46.85	57.20	0.525	0.353	2.000	
918.00	10.70	315.76	913.50	46.89	-52.39	64.65	0.884	-0.437	4.033	
999.00	8.88	319.88	993.32	57.06	-61.67	77.92	0.722	-0.674	1.526	
1,079.80	9.07	320.34	1,073.13	66.73	-69.75	89.92	0.075	0.071	0.171	
1,120.40	11.38	312.16	1,113.08	71.88	-74.76	96.90	2.012	1.707	-6.044	
1,160.80	11.86	307.61	1,152.65	77.09	-81.01	104.96	0.768	0.356	-3.379	
1,201.00	11.69	307.74	1,192.01	82.10	-87.50	113.13	0.128	-0.127	0.097	
1,241.40	12.44	309.36	1,231.51	87.37	-94.10	121.52	0.611	0.557	1.203	
1,281.80	12.06	309.62	1,270.99	92.82	-100.72	130.03	0.285	-0.282	0.193	
1,322.40	10.85	310.25	1,310.79	97.99	-106.90	138.02	0.899	-0.894	0.466	
1,335.60	10.17	309.34	1,323.76	99.53	-108.75	140.40	1.591	-1.545	-2.068	
1,387.10	7.66	329.93	1,374.65	105.39	-113.99	147.96	2.343	-1.462	11.994	
1,428.00	10.47	345.74	1,415.04	111.35	-116.27	153.07	2.736	2.061	11.597	
1,468.30	12.18	6.35	1,454.57	119.13	-116.70	157.58	3.249	1.273	15.342	
1,508.70	13.62	22.56	1,493.96	127.76	-114.40	160.24	2.881	1.069	12.037	
1,551.30	15.49	29.92	1,535.20	137.33	-109.64	161.32	1.850	1.317	5.183	
1,589.10	17.80	32.61	1,571.41	146.57	-104.01	161.49	1.932	1.833	2.135	

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Well:	F-1	North Reference:	Grid
Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
Design:	F-1	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,629.50	20.24	33.68	1,609.60	157.59	-96.80	161.27	1.830	1.812	0.795
1,663.40	23.45	33.72	1,641.07	168.08	-89.81	160.95	2.841	2.841	0.035
1,710.40	26.64	33.77	1,683.64	184.62	-78.76	160.43	2.036	2.036	0.032
1,750.30	29.38	36.44	1,718.87	199.94	-67.97	159.47	2.265	2.060	2.008
1,790.40	29.98	36.55	1,753.70	215.90	-56.16	158.00	0.451	0.449	0.082
1,831.10	30.06	36.61	1,788.94	232.25	-44.02	156.45	0.063	0.059	0.044
1,871.10	30.03	36.77	1,823.57	248.31	-32.06	154.90	0.064	-0.022	0.120
1,911.60	29.86	35.39	1,858.66	264.65	-20.15	153.55	0.526	-0.126	-1.022
1,952.20	29.78	34.81	1,893.89	281.16	-8.54	152.54	0.221	-0.059	-0.429
1,992.80	29.92	35.24	1,929.10	297.71	3.06	151.56	0.189	0.103	0.318
2,027.30	29.89	35.13	1,959.01	311.77	12.97	150.67	0.054	-0.026	-0.096
2,066.30	29.92	34.80	1,992.82	327.70	24.11	149.75	0.129	0.023	-0.254
2,113.70	29.80	35.71	2,033.92	346.98	37.73	148.51	0.297	-0.076	0.576
2,154.70	29.85	34.79	2,069.49	363.63	49.50	147.44	0.337	0.037	-0.673
2,194.30	29.84	36.98	2,103.84	379.59	61.05	146.18	0.826	-0.008	1.659
2,234.70	29.79	38.27	2,138.89	395.50	73.31	144.30	0.478	-0.037	0.958
2,275.10	30.01	37.36	2,173.92	411.41	85.66	142.34	0.374	0.163	-0.676
2,305.90	29.96	35.61	2,200.60	423.78	94.81	141.20	0.853	-0.049	-1.705
2,396.50	30.06	35.35	2,279.05	460.68	121.11	138.64	0.054	0.033	-0.086
2,436.70	30.19	35.59	2,313.82	477.12	132.82	137.51	0.132	0.097	0.179
2,470.80	30.22	36.84	2,343.29	490.96	142.95	136.32	0.554	0.026	1.100
2,517.20	30.25	36.59	2,383.38	509.69	156.92	134.49	0.084	0.019	-0.162
2,557.90	30.19	34.71	2,418.55	526.33	168.86	133.28	0.699	-0.044	-1.386
2,587.00	30.19	33.06	2,443.70	538.48	177.02	132.86	0.855	0.000	-1.701
2,653.20	27.05	19.76	2,501.87	566.63	191.21	135.87	3.211	-1.423	-6.027
2,693.70	26.44	14.67	2,538.04	584.02	196.60	140.59	1.755	-0.452	-3.770
2,733.90	26.47	6.38	2,574.04	601.59	199.87	147.20	2.754	0.022	-6.187
2,774.30	26.45	356.36	2,610.23	619.53	200.30	156.40	3.312	-0.015	-7.441
2,814.80	26.44	349.52	2,646.50	637.40	198.08	167.81	2.255	-0.007	-5.067
2,855.30	26.84	342.30	2,682.71	654.98	193.66	180.93	2.415	0.296	-5.348
2,895.10	29.13	334.29	2,717.86	672.27	186.72	196.02	3.314	1.726	-6.038
2,935.60	30.82	327.98	2,752.95	689.96	176.94	213.73	2.648	1.252	-4.674
2,975.90	33.73	322.90	2,787.03	707.64	164.72	233.51	2.960	2.166	-3.782
3,016.30	36.82	316.60	2,820.02	725.39	149.62	255.74	3.542	2.295	-4.678
3,056.60	38.42	310.99	2,851.95	742.39	131.87	279.83	2.813	1.191	-4.176
3,097.10	40.86	304.69	2,883.15	758.19	111.47	305.51	3.481	1.807	-4.667
3,136.90	45.54	301.79	2,912.15	773.09	88.67	332.74	3.831	3.528	-2.186
3,177.30	49.50	301.46	2,939.43	788.71	63.31	362.53	2.946	2.941	-0.245
3,217.80	44.76	301.22	2,966.98	804.15	37.96	392.20	3.514	-3.511	-0.178
3,257.90	40.65	301.61	2,996.44	818.32	14.76	419.39	3.081	-3.075	0.292
3,298.50	36.47	301.87	3,028.18	831.62	-6.76	444.69	3.091	-3.089	0.192
3,338.70	32.86	302.97	3,061.24	843.87	-26.07	467.55	2.734	-2.694	0.821
3,379.00	28.30	302.94	3,095.93	855.02	-43.26	488.04	3.395	-3.395	-0.022

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Well:	F-1	North Reference:	Grid
Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
Design:	F-1	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,419.20	23.61	303.38	3,132.06	864.64	-57.99	505.63	3.503	-3.500	0.328	
3,459.70	21.06	302.82	3,169.52	873.05	-70.88	521.02	1.895	-1.889	-0.415	
3,499.90	21.06	301.73	3,207.03	880.76	-83.10	535.46	0.292	0.000	-0.813	
3,540.50	21.02	301.56	3,244.93	888.41	-95.50	550.04	0.054	-0.030	-0.126	
3,580.50	20.88	301.96	3,282.28	895.93	-107.66	564.34	0.150	-0.105	0.300	
3,620.30	20.89	302.46	3,319.47	903.50	-119.67	578.53	0.135	0.008	0.377	
3,632.00	20.89	302.46	3,330.40	905.74	-123.19	582.70	0.000	0.000	0.000	

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Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
Design:	F-1	Database:	Production EDM P246N

Design Targets									
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
Volve P50 Topp - actual wellpath misses target center by 582.79m at 2765.62m MD (2602.45 TVD, 615.67 N, 200.47 E) - Point	0.00	0.94	2,859.00	96.51	266.07	6,478,660.00	435,316.00	58° 26' 33.0675 N	1° 53' 31.2283 E
Volve P70 Topp - actual wellpath misses target center by 1192.25m at 2503.67m MD (2371.69 TVD, 504.22 N, 152.86 E) - Point	0.00	0.95	2,921.90	-537.71	-29.13	6,478,026.00	435,020.90	58° 26' 12.4154 N	1° 53' 13.6817 E
Volve I20 - actual wellpath misses target center by 1499.17m at 2626.57m MD (2478.27 TVD, 555.26 N, 186.46 E) - Point	0.00	0.94	3,000.90	-765.79	665.21	6,477,798.00	435,715.00	58° 26' 5.4145 N	1° 53' 56.6978 E
Volve P51 Bunn - actual wellpath misses target center by 367.19m at 3009.51m MD (2814.57 TVD, 722.43 N, 152.38 E) - Point	0.00	0.94	3,016.02	578.68	423.63	6,479,142.00	435,473.50	58° 26' 48.7323 N	1° 53' 40.4499 E
F1 T3 040509 - actual wellpath misses target center by 2690.79m at 2653.20m MD (2501.87 TVD, 566.63 N, 191.21 E) - Point	0.00	0.00	3,054.90	-1,288.97	2,059.70	6,477,275.00	437,109.00	58° 25' 49.2378 N	1° 55' 23.1394 E
F1 TS Inj Alt 1 170409 - actual wellpath misses target center by 2820.12m at 2693.70m MD (2538.04 TVD, 584.02 N, 196.60 E) - Point	0.00	0.00	3,066.90	-966.86	2,491.85	6,477,597.00	437,541.00	58° 25' 59.8697 N	1° 55' 49.4486 E
F1 TS Inj Alt 2 170409 - actual wellpath misses target center by 2684.23m at 2596.40m MD (2451.85 TVD, 542.45 N, 179.52 E) - Point	0.00	0.00	3,066.90	-1,530.06	1,770.60	6,477,034.00	436,820.00	58° 25' 41.2973 N	1° 55' 5.5668 E
Volve I10 Topp - actual wellpath misses target center by 711.86m at 3459.70m MD (3169.52 TVD, 873.05 N, -70.88 E) - Point	0.00	0.96	3,094.90	399.62	-597.23	6,478,963.00	434,453.00	58° 26' 42.3993 N	1° 52' 37.7106 E
Volve P71 Bunn - actual wellpath misses target center by 1918.44m at 3580.50m MD (3282.28 TVD, 895.93 N, -107.66 E) - Point	0.00	0.97	3,123.70	-468.69	-1,446.72	6,478,095.00	433,603.80	58° 26' 13.8794 N	1° 51' 46.2574 E
Volve I21 Bunn - actual wellpath misses target center by 1547.34m at 2693.70m MD (2538.04 TVD, 584.02 N, 196.60 E) - Point	0.00	0.94	3,131.90	-765.79	665.21	6,477,798.00	435,715.00	58° 26' 5.4145 N	1° 53' 56.6978 E
Volve I30 Topp - actual wellpath misses target center by 1536.00m at 2814.80m MD (2646.50 TVD, 637.40 N, 198.08 E) - Point	0.00	0.93	3,133.83	-121.57	1,441.38	6,478,442.00	436,490.90	58° 26' 26.6407 N	1° 54' 43.8814 E
F1 Target 190508 - actual wellpath misses target center by 803.91m at 3557.03m MD (3260.36 TVD, 891.51 N, -100.55 E) - Point	0.00	0.00	3,151.90	407.62	-733.28	6,478,971.00	434,317.00	58° 26' 42.5844 N	1° 52' 29.3171 E
F1 DRAFT 080107 - actual wellpath misses target center by 825.19m at 3564.27m MD (3267.12 TVD, 892.88 N, -102.75 E) - Point	0.00	0.00	3,153.90	400.62	-755.28	6,478,964.00	434,295.00	58° 26' 42.3462 N	1° 52' 27.9679 E
F1 Target 080408 - actual wellpath misses target center by 835.44m at 3598.50m MD (3299.10 TVD, 899.34 N, -113.10 E) - Point	0.00	0.00	3,191.90	389.61	-766.29	6,478,953.00	434,284.00	58° 26' 41.9847 N	1° 52' 27.3010 E
F1s Target 1 080408 - actual wellpath misses target center by 2062.96m at 3632.00m MD (3330.40 TVD, 905.74 N, -123.19 E) - Point	0.00	0.00	3,207.90	-421.67	-1,697.61	6,478,142.00	433,353.00	58° 26' 15.2613 N	1° 51' 30.7483 E
F-1S T#11.1 MM 30Oct0 - actual wellpath misses target center by 365.05m at 3613.33m MD (3312.96 TVD, 902.17 N, -117.57 E) - Point	0.00	0.00	3,256.51	702.72	-418.07	6,479,266.00	434,632.10	58° 26' 52.2905 N	1° 52' 48.4421 E
F1 Pilot DRAFT 080107 - actual wellpath misses target center by 846.37m at 3632.00m MD (3330.40 TVD, 905.74 N, -123.19 E) - Point	0.00	0.00	3,264.90	838.77	-964.36	6,479,402.00	434,086.00	58° 26' 56.3915 N	1° 52' 14.6281 E

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
Design:	F-1	Database:	Production EDM P246N

F1s Target 2 080408	0.00	0.00	3,296.90	669.71	-804.30	6,479,233.00	434,246.00	58° 26' 51.0153 N	1° 52' 24.6688 E
- actual wellpath misses target center by 721.63m at 3632.00m MD (3330.40 TVD, 905.74 N, -123.19 E)									
- Point									
F1 Target 261006	0.00	0.00	4,050.90	657.71	-484.19	6,479,221.00	434,566.00	58° 26' 50.8003 N	1° 52' 44.4126 E
- actual wellpath misses target center by 843.19m at 3632.00m MD (3330.40 TVD, 905.74 N, -123.19 E)									
- Point									

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name		Casing Diameter (in)	Hole Diameter (in)
220.90	220.90	30"		30.000	36.000
1,348.40	1,336.38	20"		20.000	26.000
2,595.40	2,450.98	13 3/8"		13.375	17.500

Design Annotations				
Measured Depth (m)	Vertical Depth (m)	Local Coordinates		Comment
		+N/-S (m)	+E/-W (m)	
3,632.00	3,330.40	905.74	-123.19	Projection to TD.

Checked By: _____	Approved By: _____	Date: _____
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